

HGBS Aquila Discovery Science: Key Findings

PHASE 1: Data Exploration

- 749 cores catalogued
- 8 massive cores identified ($M > 5 \text{ Msun}$)
- 45 unusual objects discovered
- Mass range: 0.01 - 19.7 Msun

PHASE 2: Core-Filament Association

- Only 10.4% of cores on filament skeleton
- Core spacing: 0.206 pc ($\frac{1}{2}$ predicted)
- Prestellar 3× more likely on filaments
- Density drives evolution, not temperature

PHASE 3: Mass-per-unit-Length Analysis

- Median filament M_{line} : 25.8 Msun/pc
- Critical threshold CONFIRMED: 16 Msun/pc
- Prestellar M_{line} : 31.3 Msun/pc (1.45× starless)
- Protostellar M_{line} : 53.4 Msun/pc (2.5× starless)

PHASE 4: Junction Analysis

- 60 junction points identified
- 1,603 high-density zones (convergence zones)
- Environmental mass scaling: 3.4× on filaments
- Massive cores NOT on junctions (revised model)

PHASE 5: Discovery Mode with ASTRA

- 60 anomalous objects detected
- 4 new scientific hypotheses generated
- Local density CAUSES evolution ($p < 10^{-34}$)
- Mass- α_{BE} correlation: $r = -0.30$

All findings support Two-Stage Evolution Model:

Stage 1: Mass accumulation in filamentary environments

Stage 2: Gravitational collapse when critical density/ M_{line} is reached